Exam Date & Time: 20-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

III SEMESTER B.TECH. END SEMESTER EXAMINATION

ENGINEERING MATHEMATICS - III [MAT 2121]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) Solve the boundary value problem using finite difference method:
$y'' - 64y + 10 = 0$, $y(0) = 0,\ y(1) = 0$ with $h = \frac{1}{4}$. (4)

1B) Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for $0 < x < 1,\ \ 0 < y < 1$, given that $h = \frac{1}{3}, u(x, 0) = 0 = u(0, y), u(x, 1) = 6x$ and $u(1, y) = 3y$.. (3)

1C) Solve $4\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ subject to $u(0,t) = u(4,t) = 0, u_t(x,0) = 0$ and $u(x,0) = x(4-x)$ by taking $h = 1,$ up to 4-time steps. (3)

2A) Compute u for four time steps with, $h = 0.25,$ the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$,
$0 < x < 1,\ \ t > 0,\ \lambda = \frac{1}{2}, u(x,0) = 100\ (x - x^2)$,
$u(0,t) = u(1,t) = 0$. (4)

2B) Find the constant term and the first harmonic of the Fourier series from the following data.

x	0	2	4	6	8	10	12
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0

(3)

2C) Obtain half range cosine and sine series for $f(x) = x$ in the interval $0 \le x \le \pi$. (3)

3A) Obtain the Fourier series expansion for the function

$$f(x) = \begin{cases} 0 & , \quad -\pi \le x \le 0 \\ \frac{1}{4}\pi x & , \quad 0 \le x \le \pi \end{cases}$$

(4)

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$

and hence deduce that .

3B) Find the Fourier transform of the function $f(x) = \begin{cases} 1 - |x|, & for\ |x| \le 1 \\ 0, & for\ |x| > 1 \end{cases}$. Hence deduce $\int_0^\infty \left(\frac{\sin t}{t}\right)^2 dt = \frac{\pi}{2}$. (3)

3C) Find the equations for the tangent plane and normal line to the surface $xz^2 + x^2y = z - 1$ at (1, -3, 2). (3)

4A) Find the value of $'a'$, such that $\vec{F} = (axy - z^3)i + (a-2)x^2j + (1-a)xz^2k$ is irrotational and find the scalar function ϕ such that $\nabla \phi = \vec{F}$. (4)

4B) By using divergence theorem, evaluate $\iint_S \vec{f} \cdot \hat{n}\, ds$, where $\vec{f} = 4x\hat{\imath} - 2y^2\hat{\jmath} + z^2\,\hat{k}$ and S is the surface enclosing the region $x^2 + y^2 = 9$ and $z = 0\ and\ z = 4$. (3)

4C) Suppose $\vec{F} = (3x^2 + 6y)\hat{\imath} - 14yz\hat{\jmath} + 20xz^2\hat{k}$. Evaluate $\int_C \vec{F}.d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the path $C: x = t, y = t^2, z = t^3$. (3)

5A) Verify Green's theorem for $\oint_C (xy^2 + y^2)dx + x^2dy$ where C is the curve bounded by $y = x$ and $y = x^2$. . (4)

5B) Solve $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$ by the method of separation of variables. (3)

5C) Using the solution of D'Alembert's wave equation, solve for u, when $u(x,0) = f(x)$ and $\frac{\partial u}{\partial t}(x,0) = 0$. (3)

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